

Section 2: ROS Workspace and Programming Basics

AA174A - Principles of Robot Autonomy I

Week 2 – Autumn 2026

Section Goals

- Start with **mini-lecture**.
- Work through tasks in write-up with groups of 3-4.
- Implement and **test code from HW on Turtlebots (sim)**.
- You can leave when a CA signs off on all of your tasks!



ROS



Initially released in 2007 by the founder of Zipline!
16+ years of open-source.

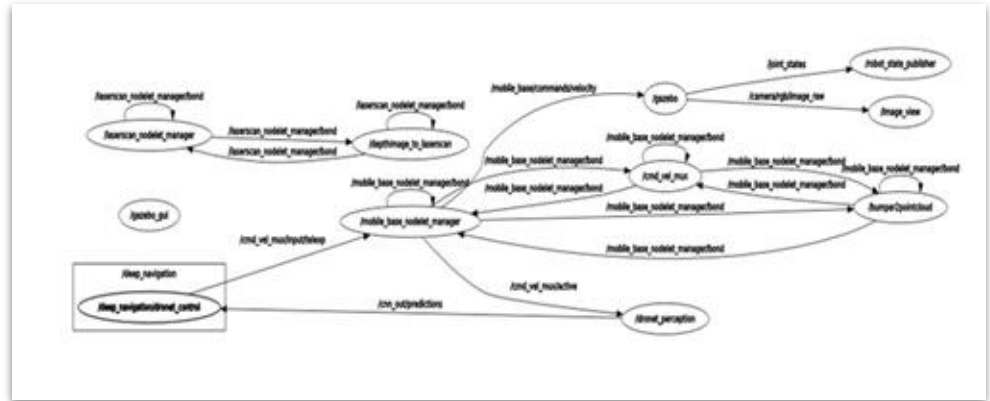
What is ROS?

- **Event programming, networking, and build tools.**
- **Drivers** for sensors and motors.
- **Visualization** and logging tools.
- An interface for all sorts of **external libraries** (SLAM, planning, perception).



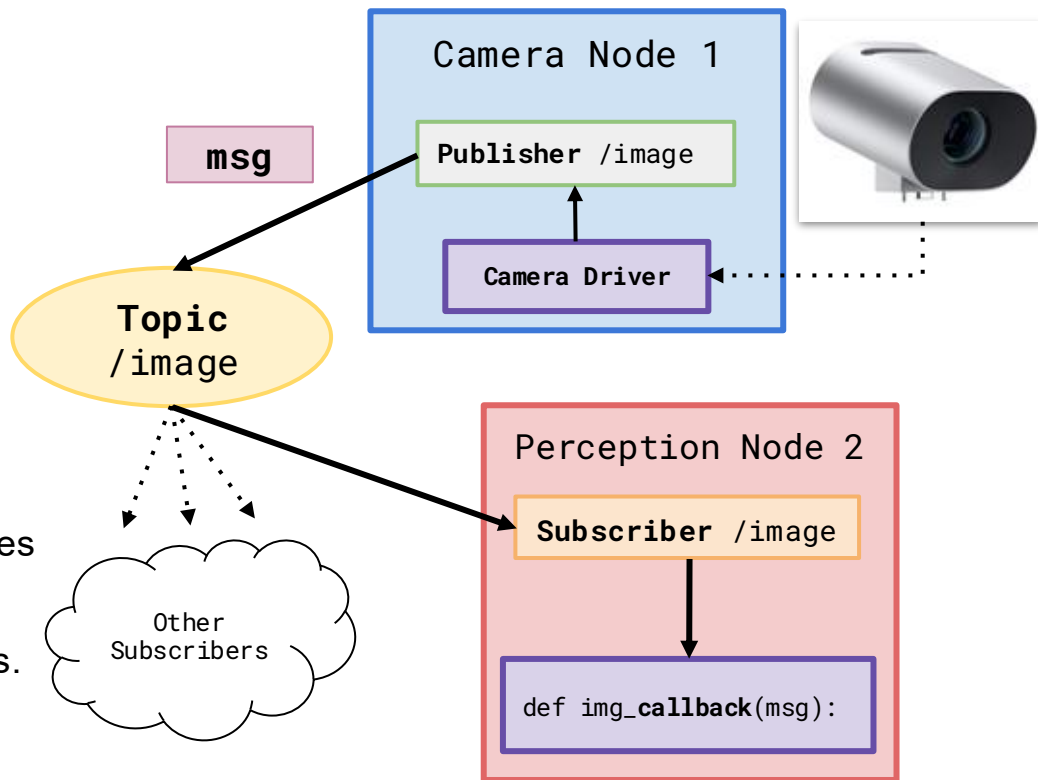
Why use event programming?

- **Even simple robots are complicated!**
- Lots of sensors and actuators means that we need **scalable tools** to handle everything.
- Event programming is a scalable paradigm



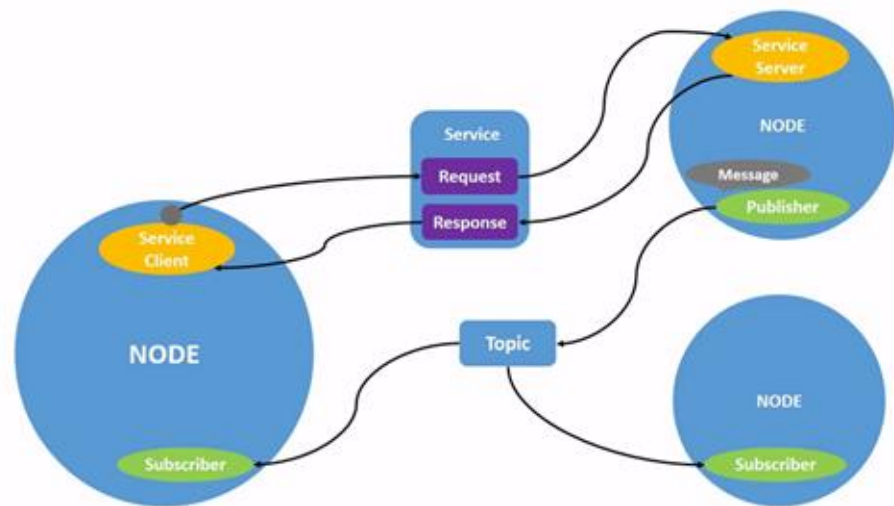
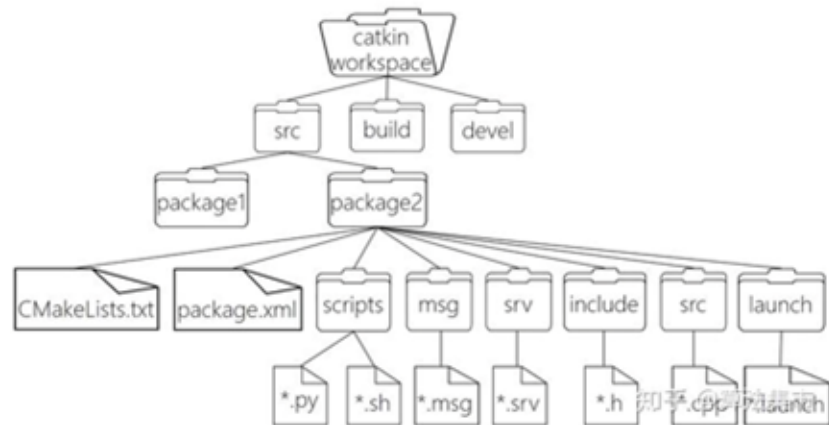
ROS Event Programming Basics

- Nodes
 - Manage **publishers/subscribers**, **callbacks**, and external code.
- Topics and Messages
 - Topics are “hubs” where messages are “posted” by publishers.
 - **Messages are formatted data!**
- Publishers
 - Send messages **to** topics **from** nodes
- Subscribers
 - Pull messages **from** topics **to** nodes.



ROS Communication

- Message Types
 - Data structure that holds some information about the robot
- Publication
 - Broadcast message to the ROS network
- Subscription
 - Listens to some broadcasted channel



ROS Development Environment

- **Workspaces**
 - Organizes packages.
- **Packages**
 - Organizes code for nodes.
- **CMakeLists.txt**
 - Tells *colcon* how to build.
- **package.xml**
 - Tells *rosdep* what to install.

```
~/tb_ws/      # My group's workspace root directory
build/        # Compiled libraries
install/setup.bash # Source workspace packages
src/          # Top-level package directory
  external_pkg1/ # Some external package that I am using
  external_pkg2/ # Another external package
  group4_repo/   # My group's autonomy repository
    navigation_pkg/ # My group's Navigation ROS Package
    perception_pkg/ # My group's Perception ROS Package
      CMakeLists.txt # colcon build instructions
      package.xml    # ROS package details
      src/           # C++ Code lives here!
  my_pytorch_py_lib/ # My custom python module
  scripts/          # ROS Python Executables
    camera_node.py   # Camera node code
    detection_node.py # Detection node code
```

Live ROS Demo.